

```
// ===== SNAPSHOT DATA (2026-06-29) - used as offline-first base
=====
const SNAPSHOT = [
  { id:'bitcoin', sym:'BTC', name:'Bitcoin', price:59668.09, ch24:-0.43, ch7:-2.1,
mcap:1209531900520, vol:31512449517 },
  { id:'ethereum', sym:'ETH', name:'Ethereum', price:1586.26, ch24:0.39, ch7:1.8,
mcap:195520777963, vol:12454909904 },
  { id:'binancecoin', sym:'BNB', name:'BNB', price:554.17, ch24:-0.18, ch7:0.6,
mcap:75524653023, vol:765022148 },
  { id:'ripple', sym:'XRP', name:'XRP', price:1.05, ch24:-0.31, ch7:0.98,
mcap:66388781741, vol:1938176312 },
  { id:'solana', sym:'SOL', name:'Solana', price:74.25, ch24:2.04, ch7:4.97,
mcap:43799746717, vol:4229713062 },
  { id:'dogecoin', sym:'DOGE', name:'Dogecoin', price:0.07248, ch24:-1.15,
ch7:-3.6, mcap:10670000000, vol:980000000 },
  { id:'tron', sym:'TRX', name:'TRON', price:0.3215, ch24:0.26, ch7:-3.6,
mcap:30474299007, vol:575108814 },
  { id:'cardano', sym:'ADA', name:'Cardano', price:0.398, ch24:-0.8, ch7:-1.2,
mcap:14200000000, vol:320000000 },
  { id:'polkadot', sym:'DOT', name:'Polkadot', price:4.12, ch24:3.4, ch7:8.1,
mcap:6100000000, vol:210000000 },
  { id:'avalanche-2', sym:'AVAX', name:'Avalanche', price:18.6, ch24:1.1, ch7:2.3,
mcap:7600000000, vol:280000000 },
  { id:'chainlink', sym:'LINK', name:'Chainlink', price:11.4, ch24:0.5, ch7:-0.9,
mcap:7300000000, vol:340000000 },
  { id:'litecoin', sym:'LTC', name:'Litecoin', price:78.3, ch24:-0.6, ch7:1.4,
mcap:5900000000, vol:410000000 },
  { id:'shiba-inu', sym:'SHIB', name:'Shiba Inu', price:0.0000112, ch24:-1.8,
ch7:-4.2, mcap:6600000000, vol:190000000 },
  { id:'uniswap', sym:'UNI', name:'Uniswap', price:6.8, ch24:1.9, ch7:3.1,
mcap:4100000000, vol:120000000 },
  { id:'near', sym:'NEAR', name:'NEAR Protocol', price:2.9, ch24:2.6, ch7:5.4,
mcap:3300000000, vol:140000000 },
  { id:'aptos', sym:'APT', name:'Aptos', price:4.4, ch24:-1.1, ch7:-2.8,
mcap:2500000000, vol:95000000 },
  { id:'arbitrum', sym:'ARB', name:'Arbitrum', price:0.42, ch24:0.8, ch7:1.6,
mcap:1700000000, vol:88000000 },
  { id:'optimism', sym:'OP', name:'Optimism', price:0.75, ch24:-0.4, ch7:0.3,
mcap:1300000000, vol:62000000 },
  { id:'sui', sym:'SUI', name:'Sui', price:1.85, ch24:3.9, ch7:7.2,
mcap:5400000000, vol:310000000 },
  { id:'stellar', sym:'XLM', name:'Stellar', price:0.1723, ch24:-0.9, ch7:-1.6,
mcap:5200000000, vol:140000000 },
];
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const GLOBAL_SNAPSHOT = { totalMcap: 2.2e12, totalVol: 82e9, btcDom: 56.0, ethDom:
8.9 };

const NEWS_SNAPSHOT = [
  { tag:'市場', title:'比特幣主導率守在 56-60% 區間，山寨幣季節指數約 46，仍屬比特幣領漲格
局', time:'今天' },
  { tag:'ETF', title:'現貨比特幣 ETF 累積流入已突破 569 億美元，機構資金持續鎖定在 BTC 生
態', time:'今天' },
  { tag:'山寨幣', title:'分析師指出需確認跌破 55% 主導率才是真正資金輪動訊號，目前已超過 260
天未見山寨幣季', time:'1天前' },
  { tag:'監管', title:'英國 FCA 公布加密貨幣公司新規定，最終期限訂在 2027 年 2 月',
time:'2天前' },
  { tag:'生態', title:'Solana 基金會推出 STRIDE 安全框架與 Alpenglow 共識升級提案',
time:'3天前' },
];

// ===== STATE (persisted via localStorage in real PWA context)
=====
let coins = JSON.parse(JSON.stringify(SNAPSHOT));
let globalStats = { ...GLOBAL_SNAPSHOT };
let dataIsLive = false;
let currentDetailId = 'bitcoin';
let currentTF = 7;

function loadState(key, fallback) {
  try {
    const v = localStorage.getItem(key);
    return v ? JSON.parse(v) : fallback;
  } catch(e) { return fallback; }
}

function saveState(key, val) {
  try { localStorage.setItem(key, JSON.stringify(val)); } catch(e) {}
}

let favorites = loadState('cai_favorites', ['bitcoin','ethereum','solana']);
let holdings = loadState('cai_holdings', []); // [{id, amount}]
let alerts = loadState('cai_alerts', []); // [{id, coinId, cond, price, fired}]

// ===== UTIL =====
function fmtPrice(p) {
  if (p == null) return '--';
  if (p >= 1000) return '$' + p.toLocaleString('en-US',{maximumFractionDigits:0});
  if (p >= 1) return '$' + p.toLocaleString('en-US',
{minimumFractionDigits:2,maximumFractionDigits:2});
  if (p >= 0.01) return '$' + p.toFixed(4);
  return '$' + p.toFixed(8);
}

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}
function fmtBig(n) {
  if (!n) return '--';
  if (n >= 1e12) return '$'+(n/1e12).toFixed(2)+'T';
  if (n >= 1e9) return '$'+(n/1e9).toFixed(1)+'B';
  if (n >= 1e6) return '$'+(n/1e6).toFixed(0)+'M';
  return '$'+n.toFixed(0);
}

function getCoin(id) { return coins.find(c => c.id === id); }
function toast(msg) {
  const t = document.getElementById('toast');
  t.textContent = msg;
  t.classList.add('show');
  setTimeout(() => t.classList.remove('show'), 2200);
}

// ===== SIGNAL LOGIC =====
function computeSignal(c) {
  const ch = c.ch24, ch7 = c.ch7;
  if (ch > 4 && ch7 > 5) return 'BUY';
  if (ch < -4 && ch7 < -5) return 'SELL';
  if (ch < -2 && ch7 < -8) return 'SELL';
  if (ch > 2 && ch7 > 0) return 'BUY';
  return 'HOLD';
}

function signalLabel(s) { return s==='BUY' ? '買入' : s==='SELL' ? '賣出' : '觀望'; }

function signalReason(c) {
  const s = computeSignal(c);
  if (s === 'BUY') return `24h ${c.ch24>=0?'+'':''}${c.ch24.toFixed(1)}%, 7d
${c.ch7>=0?'+'':''}${c.ch7.toFixed(1)}%, 動能轉強，短線偏多排列。`;
  if (s === 'SELL') return `24h ${c.ch24.toFixed(1)}%, 7d ${c.ch7.toFixed(1)}%, 賣
壓明顯，留意支撐是否跌破。`;
  return `24h ${c.ch24>=0?'+'':''}${c.ch24.toFixed(1)}%, 盤整格局，建議觀望等待方向確認。
`;
}

// ===== RENDER: TOP TICKER =====
function renderTicker() {
  const top6 = coins.slice(0, 6);
  document.getElementById('ticker-strip').innerHTML = top6.map(c => {
    const dir = c.ch24 >= 0 ? 'up' : 'dn';
    return `${c.sym} <b>${fmtPrice(c.price)}</b> <span
class="${dir}">${c.ch24>=0?'+'':''}${c.ch24.toFixed(1)}%</span></span>`;
  }).join('');
}

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// ===== RENDER: COIN ROW =====
function coinRowHTML(c, showStar=true) {
  const dir = c.ch24 >= 0 ? 'up' : 'dn';
  const sig = computeSignal(c);
  const isFav = favorites.includes(c.id);
  return `

${c.sym.slice(0,4)}</div>`
    `

${c.sym} <span class="sig-tag`
    `${sig}">${signalLabel(sig)}</span></div>`
      `


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    const btn = document.getElementById('fav-toggle');
    const isFav = favorites.includes(currentDetailId);
    btn.textContent = isFav ? '★ 已加入最愛' : '☆ 加入最愛';
    btn.classList.toggle('active', isFav);
}

// ===== NAVIGATION =====
function switchScreen(name) {
    document.querySelectorAll('.screen').forEach(s => s.classList.remove('active'));
    document.getElementById('screen-'+name).classList.add('active');
    document.querySelectorAll('.nav-item').forEach(n => n.classList.toggle('active',
n.dataset.screen === name));
    if (name === 'portfolio') renderPortfolio();
    if (name === 'alerts') { renderAlerts(); renderNews(); }
}

function openDetail(id) {
    currentDetailId = id;
    switchScreen('detail-internal'); // placeholder, handled below
    document.querySelectorAll('.screen').forEach(s => s.classList.remove('active'));
    document.getElementById('screen-detail').classList.add('active');
    renderDetail();
}

// ===== SIMPLE PSEUDO-CANDLE GENERATION (offline, deterministic from
price) =====
function generateSeries(c, days) {
    const points = days === 1 ? 24 : days === 7 ? 42 : days === 30 ? 60 : 90;
    const totalChange = days === 1 ? c.ch24 : days === 7 ? c.ch7 : days === 30 ?
c.ch7*1.8 : c.ch7*3.2;
    const startPrice = c.price / (1 + totalChange/100);
    let seed = c.sym.charCodeAt(0) * 7 + c.sym.charCodeAt(1) * 13;
    function rand() { seed = (seed * 9301 + 49297) % 233280; return seed / 233280; }
    const series = [];
    let p = startPrice;
    for (let i=0;i<points;i++) {
        const progress = i/(points-1);
        const target = startPrice + (c.price - startPrice) * progress;
        const noise = (rand()-0.5) * Math.abs(c.price)*0.015;
        p = target + noise;
        series.push(p);
    }
    series[series.length-1] = c.price;
    return series;
}

// ===== INDICATORS =====

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function calcRSI(closes, period=14) {
  if (closes.length < period+1) period = Math.max(3, closes.length-2);
  let gains=0, losses=0;
  for (let i=1;i<=period;i++) {
    const d = closes[i]-closes[i-1];
    if (d>=0) gains+=d; else losses-=d;
  }
  let avgGain=gains/period, avgLoss=losses/period;
  for (let i=period+1;i<closes.length;i++) {
    const d = closes[i]-closes[i-1];
    const g = d>0?d:0, l=d<0?-d:0;
    avgGain = (avgGain*(period-1)+g)/period;
    avgLoss = (avgLoss*(period-1)+l)/period;
  }
  if (avgLoss===0) return 100;
  const rs = avgGain/avgLoss;
  return 100 - (100/(1+rs));
}

function ema(values, period) {
  const k = 2/(period+1);
  let out = [values[0]];
  for (let i=1;i<values.length;i++) out.push(values[i]*k + out[i-1]*(1-k));
  return out;
}

function calcMACD(closes) {
  const p = Math.min(12, Math.floor(closes.length/3));
  const p2 = Math.min(26, Math.floor(closes.length/1.5));
  if (closes.length < 5) return null;
  const e1 = ema(closes, Math.max(3,p)), e2 = ema(closes, Math.max(5,p2));
  const macdLine = e1.map((v,i)=>v-e2[i]);
  const sig = ema(macdLine, 5);
  return { hist: macdLine[macdLine.length-1] - sig[sig.length-1] };
}

function calcMA(closes, period) {
  const p = Math.min(period, closes.length);
  return closes.slice(-p).reduce((a,b)=>a+b,0)/p;
}

function calcATR(closes) {
  if (closes.length<2) return 0;
  let sum=0;
  for (let i=1;i<closes.length;i++) sum += Math.abs(closes[i]-closes[i-1]);
  return (sum/(closes.length-1))/closes[closes.length-1]*100;
}

// ===== MINI CHART (canvas) =====
function drawChart(series, up) {
  const canvas = document.getElementById('mini-chart');

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const dpr = window.devicePixelRatio || 1;
const rect = canvas.getBoundingClientRect();
canvas.width = rect.width * dpr;
canvas.height = 160 * dpr;
const ctx = canvas.getContext('2d');
ctx.scale(dpr, dpr);
const w = rect.width, h = 160;
ctx.clearRect(0,0,w,h);

const min = Math.min(...series), max = Math.max(...series);
const range = (max-min) || 1;
const pad = 8;

const color = up ? '#2dd4a3' : '#ff5c6c';

// gradient fill
const grad = ctx.createLinearGradient(0,0,0,h);
grad.addColorStop(0, up ? 'rgba(45,212,163,0.25)' : 'rgba(255,92,108,0.25)');
grad.addColorStop(1, 'rgba(0,0,0,0)');

ctx.beginPath();
series.forEach((v,i) => {
  const x = (i/(series.length-1)) * w;
  const y = pad + (1 - (v-min)/range) * (h - pad*2);
  if (i===0) ctx.moveTo(x,y); else ctx.lineTo(x,y);
});
ctx.lineTo(w, h); ctx.lineTo(0, h); ctx.closePath();
ctx.fillStyle = grad; ctx.fill();

ctx.beginPath();
series.forEach((v,i) => {
  const x = (i/(series.length-1)) * w;
  const y = pad + (1 - (v-min)/range) * (h - pad*2);
  if (i===0) ctx.moveTo(x,y); else ctx.lineTo(x,y);
});
ctx.strokeStyle = color; ctx.lineWidth = 2; ctx.lineJoin='round'; ctx.stroke();
}

// ===== DETAIL SCREEN =====
function renderDetail() {
  const c = getCoin(currentDetailId);
  if (!c) return;
  document.getElementById('d-sym').textContent = `${c.sym} ·
  ${c.name.toUpperCase()}`;
  document.getElementById('d-price').textContent = fmtPrice(c.price);
  const chgEl = document.getElementById('d-chg');
  chgEl.textContent = `${c.ch24>=0?'+' ':''}${c.ch24.toFixed(2)}% (24h)`;

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    chgEl.style.color = c.ch24>=0 ? 'var(--up)' : 'var(--down)';

    updateFavToggleBtn();
    renderChartAndIndicators();
    renderAIForDetail(c);
}

function renderChartAndIndicators() {
    const c = getCoin(currentDetailId);
    const series = generateSeries(c, currentTF);
    drawChart(series, c.ch24 >= 0);

    const rsi = calcRSI(series);
    const rsiEl = document.getElementById('ind-rsi'), rsiFill =
document.getElementById('ind-rsi-fill'), rsiCap = document.getElementById('ind-
rsi-cap');
    rsiEl.textContent = rsi.toFixed(1);
    rsiFill.style.width = rsi+'%';
    if (rsi>70) { rsiEl.className='ind-num dn'; rsiFill.style.background='var(--
down)'; rsiCap.textContent='超買區間'; }
    else if (rsi<30) { rsiEl.className='ind-num up'; rsiFill.style.background='var(-
up)'; rsiCap.textContent='超賣區間'; }
    else { rsiEl.className='ind-num neu'; rsiFill.style.background='var(--accent)';
rsiCap.textContent='中性區間'; }

    const macd = calcMACD(series);
    const macdEl = document.getElementById('ind-macd'), macdCap =
document.getElementById('ind-macd-cap');
    if (macd) {
        macdEl.textContent = macd.hist>0?'正向':'負向';
        macdEl.className = 'ind-num ' + (macd.hist>0?'up':'dn');
        macdCap.textContent = macd.hist>0 ? '柱狀圖轉正，動能轉強' : '柱狀圖轉負，動能偏弱';
    }

    const ma7 = calcMA(series,7), maLong = calcMA(series,
Math.min(25,series.length));
    const maEl = document.getElementById('ind-ma'), maCap =
document.getElementById('ind-ma-cap');
    const bullish = ma7 > maLong;
    maEl.textContent = bullish ? '偏多' : '偏空';
    maEl.className = 'ind-num ' + (bullish?'up':'dn');
    maCap.textContent = bullish ? '短均線在長均線之上' : '短均線在長均線之下';

    const atr = calcATR(series);
    const atrEl = document.getElementById('ind-atr'), atrCap =
document.getElementById('ind-atr-cap');
    atrEl.textContent = atr.toFixed(2)+'%';

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atrEl.className = 'ind-num ' + (atr>4?'dn':atr>1.5?'neu':'up');
atrCap.textContent = atr>4 ? '波動劇烈' : atr>1.5 ? '正常區間' : '波動平緩';

// signal banner
const sig = computeSignal(c);
const banner = document.getElementById('signal-banner');
banner.className = 'signal-banner ' + sig;
document.getElementById('sig-verdict').textContent = (sig==='BUY'?'+ 買入訊號':sig==='SELL'?'+ 賣出訊號':'- 觀望訊號');
document.getElementById('sig-why').textContent = signalReason(c) + `RSI ${rsi.toFixed(0)}, 均線${bullish?'偏多':'偏空'}排列, 波動率${atr.toFixed(1)}%。`;
}

document.getElementById('tf-row').addEventListener('click', (e) => {
  const pill = e.target.closest('.tf-pill');
  if (!pill) return;
  document.querySelectorAll('.tf-pill').forEach(p=>p.classList.remove('active'));
  pill.classList.add('active');
  currentTF = parseInt(pill.dataset.d);
  renderChartAndIndicators();
});

function renderAIForDetail(c) {
  const sig = computeSignal(c);
  const verdict = sig==='BUY' ? '技術面與動能呈現偏多訊號' : sig==='SELL' ? '短線賣壓較重, 建議謹慎' : '目前處於盤整格局, 方向尚不明確';
  const strategy = sig==='BUY'
    ? '短線可關注回踩支撐的進場機會; 中長期持有者可考慮分批加碼。'
    : sig==='SELL'
    ? '短線交易者宜降低曝險或設定停損; 中長期持有者可觀察是否跌破關鍵支撐再決定加碼時機。'
    : '建議耐心等待量價配合的突破訊號, 避免在盤整區間頻繁進出。';

  document.getElementById('d-ai-report').innerHTML = `
    <h3>市場研判</h3>
    <p>${c.name} (${c.sym}) 24小時${c.ch24>=0?'上漲':'下跌'}
    ${Math.abs(c.ch24).toFixed(2)}%, 7日${c.ch7>=0?'上漲':'下跌'}
    ${Math.abs(c.ch7).toFixed(2)}%。${verdict}</p>
    <h3>操作策略</h3>
    <p>${strategy}</p>
    <h3>風險提示</h3>
    <p>加密貨幣價格波動劇烈, 本分析基於歷史價格動能計算, 不構成投資建議, 請自行評估風險並謹慎操作。</p>
  `;
}

// ===== PORTFOLIO =====
function populateCoinSelect(selectEl) {

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    selectEl.innerHTML = coins.map(c => `

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        <div class="pf-del" onclick="removeHolding('${h.id}')">✕</div>
    </div>`;
    }).join('');

    document.getElementById('portfolio-list').innerHTML = rows || `<div
class="empty-state"><div class="icon">◆</div>尚未新增任何持倉</div>`;
    document.getElementById('pf-total').textContent = fmtPrice(total);
    const avgCh = total > 0 ? weightedCh/total : 0;
    const chgEl = document.getElementById('pf-total-chg');
    if (total > 0) {
        chgEl.textContent = `${avgCh>=0?'+' : ''}${avgCh.toFixed(2)}% (24h 加權)`;
        chgEl.style.color = avgCh>=0 ? 'var(--up)' : 'var(--down)';
    } else {
        chgEl.textContent = '尚無持倉';
        chgEl.style.color = 'var(--text-faint)';
    }
}

// ===== ALERTS =====
function openAlertModal() {
    populateCoinSelect(document.getElementById('a-coin'));
    if (currentDetailId) document.getElementById('a-coin').value = currentDetailId;
    document.getElementById('a-price').value = '';
    showModal('modal-alert');
}

function confirmAddAlert() {
    const id = document.getElementById('a-coin').value;
    const cond = document.getElementById('a-cond').value;
    const price = parseFloat(document.getElementById('a-price').value);
    if (!price || price <= 0) { toast('請輸入有效價格'); return; }
    alerts.push({ id: Date.now().toString(), coinId: id, cond, price, fired:false
});
    saveState('cai_alerts', alerts);
    closeModal('modal-alert');
    renderAlerts();
    toast('提醒已設定');
}

function removeAlert(id) {
    alerts = alerts.filter(a => a.id !== id);
    saveState('cai_alerts', alerts);
    renderAlerts();
}

function renderAlerts() {
    const rows = alerts.map(a => {
        const c = getCoin(a.coinId);
        if (!c) return '';
        return `<div class="pf-row">

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<div class="coin-glyph">${c.sym.slice(0,4)}</div>
<div class="pf-row-info">
  <div class="coin-sym">${c.sym}</div>
  <div class="pf-row-amt">${a.cond==='above'?'高於':'低於'}
${fmtPrice(a.price)}</div>
</div>
<div class="pf-row-val">${a.fired ? '<span style="color:var(--accent)">已觸發
</span>' : '監控中'}</div>
<div class="pf-del" onclick="removeAlert('${a.id}')">x</div>
</div>`;
}).join('');
document.getElementById('alerts-list').innerHTML = rows || `<div class="empty-
state"><div class="icon">🕒</div>尚未設定任何提醒</div>`;
}
function checkAlerts() {
  let changed = false;
  alerts.forEach(a => {
    if (a.fired) return;
    const c = getCoin(a.coinId);
    if (!c) return;
    if ((a.cond==='above' && c.price >= a.price) || (a.cond==='below' && c.price
<= a.price)) {
      a.fired = true;
      changed = true;
      toast(`🔔 ${c.sym} 已${a.cond==='above'?'突破':'跌破'} ${fmtPrice(a.price)}`);
      if ('Notification' in window && Notification.permission === 'granted') {
        try { new Notification(`${c.sym} 價格提醒`, { body: `已
${a.cond==='above'?'突破':'跌破'} ${fmtPrice(a.price)}` }); } catch(e){}
      }
    }
  });
  if (changed) { saveState('cai_alerts', alerts); renderAlerts(); }
}

// ===== NEWS =====
function renderNews() {
  document.getElementById('news-list').innerHTML = NEWS_SNAPSHOT.map(n => `
<div class="news-card">
  <div class="news-tag">${n.tag}</div>
  <div class="news-title">${n.title}</div>
  <div class="news-meta">${n.time}</div>
</div>
`).join('');
}

// ===== MODAL =====
function showModal(id) { document.getElementById(id).classList.add('show'); }

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function closeModal(id) { document.getElementById(id).classList.remove('show'); }
document.querySelectorAll('.modal-overlay').forEach(m => {
  m.addEventListener('click', (e) => { if (e.target === m)
m.classList.remove('show'); });
});

// ===== LIVE DATA FETCH (best-effort, never blocks) =====
async function tryFetchLive() {
  const statusPill = document.getElementById('status-pill');
  const statusText = document.getElementById('status-text');
  const ids = coins.map(c=>c.id).join(',');
  const controller = new AbortController();
  const timeout = setTimeout(()=>controller.abort(), 6000);
  try {
    const res = await fetch(`https://api.coingecko.com/api/v3/coins/markets?
vs_currency=usd&ids=${ids}&price_change_percentage=7d`, { signal:
controller.signal });
    clearTimeout(timeout);
    if (!res.ok) throw new Error('bad response');
    const data = await res.json();
    if (!Array.isArray(data) || !data.length) throw new Error('empty');
    const byId = {};
    data.forEach(d => byId[d.id] = d);
    coins.forEach(c => {
      const d = byId[c.id];
      if (!d) return;
      c.price = d.current_price;
      c.ch24 = d.price_change_percentage_24h || 0;
      c.ch7 = d.price_change_percentage_7d_in_currency || 0;
      c.mcap = d.market_cap;
      c.vol = d.total_volume;
    });
    dataIsLive = true;
    statusPill.className = 'status-pill live';
    statusText.textContent = '即時數據 · ' + new Date().toLocaleTimeString('zh-TW',
{hour12:false});
    refreshAllViews();
  } catch(e) {
    dataIsLive = false;
    statusPill.className = 'status-pill stale';
    statusText.textContent = '離線快照數據（無法連線）';
  }
}

function manualRefresh() { toast('正在嘗試更新...'); tryFetchLive(); }

function refreshAllViews() {

```

```

    renderTicker();
    renderMarketScreen();
    if (document.getElementById('screen-detail').classList.contains('active'))
renderDetail();
    if (document.getElementById('screen-portfolio').classList.contains('active'))
renderPortfolio();
    checkAlerts();
}

// ===== CLOCK =====
setInterval(() => {
    document.getElementById('clock').textContent = new
Date().toLocaleTimeString('zh-TW',{hour12:false});
}, 1000);

// ===== NOTIFICATION PERMISSION (best-effort) =====
if ('Notification' in window && Notification.permission === 'default') {
    // ask politely after first interaction, not on load
}

// ===== SERVICE WORKER =====
if ('serviceWorker' in navigator) {
    window.addEventListener('load', () => {
        navigator.serviceWorker.register('sw.js').catch(()=>{});
    });
}

// ===== INIT =====
renderTicker();
renderMarketScreen();
renderDetail();
document.getElementById('status-text').textContent = '離線快照數據 (2026/6/29) ';
tryFetchLive();

```